

What is HTTP(S)?

HTTP stands for Hyper Text Transfer protocol. It is what used whenever you view a website, developed by Tim Berners-Lee and his team b/w 1989-1991.

* HTTP is set of rules used for communicating with web servers for transmitting of web page data, whether that is HTML, images, video etc.

(means HTTP is official language your web browser used to ask websites computer for its text, pictures, and videos so it can display them on your screen).

What is HTTPS? (Hyper Text Transfer protocol Secure)

* HTTPS is secure version of HTTP. HTTPS data is encrypted so it not only stops people from seeing the data you are receiving and sending. But it also gives you assurance that you're talking to correct web server and not something impersonating it.

Quiz:-

1) What does HTTP stand for? 2) What does S in HTTPS stand for?

A) Hyper Text Transfer protocol A) Secure

Requests and Responses :- Task 2

* To access a website, your browser will need to make requests to web server for assets such as HTML, images and download the responses. Before that you need to tell browser specifically how and where to access these resources, this is where URL's will help.

Note:- Website means websites like toshakme, linkedin, Encyclopedia.

Browser means like Google, Chrome, Microsoft edge and Safari.

URL:- URL stands for Uniform Resource Locator. If you've used the internet, you've used URL before. A URL is predominantly an instruction on how to access a resource on the internet. URL looks like below image

http://user:password@toshakme.com:80/video-game?id=1#task2

Labels:
scheme: http
Host/domain: toshakme.com
Port: 80
Path: /video-game
Fragment: #task2

↑
user

↑
port

↑
Query string

Scheme: This instructs on what protocol to use for accessing resource such as HTTP, HTTPS, FTP.

user: - Some services requires authentication to log in, you can put username and password into URL to log in

Host: - The domain name or IP addresses of server you wish to access.

port: - The port you are going to connect, 80 for HTTP and 443 for HTTPS, but this can be hosted on any port b/w 1-65535.

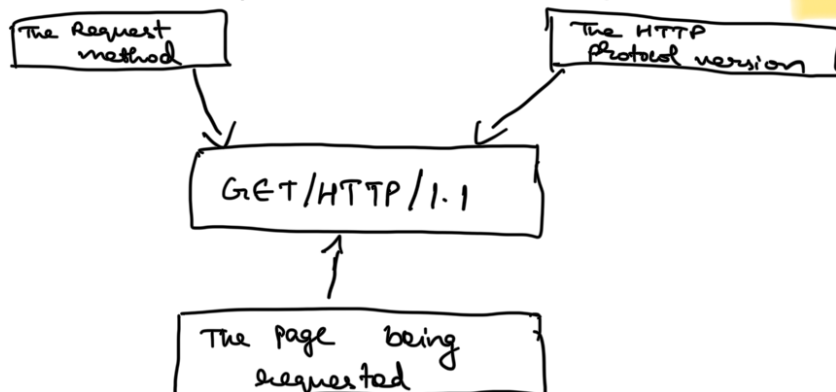
path: - The file name or location you are trying to access.

QueryString: - Extra bit of information that can be sent to requested path. Ex, blog?id=1 would tell blog path that you wish to receive blog article with id of 1.

Fragment: - This is a reference of location on actual page requested. This is commonly used for pages with long content and can have certain part of page directly to linked to it, so it is viewable to user as soon as they access the page.

Making a Request: -

It is possible to make request to web server with just one line **GET /HTTP /1.1**



But for much richer web experience, you'll need to send other data as well. This other data is sent in what is called **headers**, where headers contain extra information to give to the web server you're communicating with, but we'll get more into this in header task.

Example request: -

Get/HTTP/1.1

Host: toyhackme.com

User-Agent: Mozilla/5.0 Firefox/87.0

Referer: - https://toyhackme.com/

To break down each line of request

link 1:- The request is sending the GET method (more on this in HTTP methods task), request home page with / and telling web servers we are using HTTP protocol version 1.1

link 2:- we tell all web server we want the website Toyhackme.com

link 3:- we tell web server we are using Firefox version 87 Browser.

link 4:- we are telling web server that webpage that referred us to this one is https://toyhackme.com

link 5:- HTTP requests always ends with blank line to inform web server that request has finished.

Example Response:-

HTTP/1.1 200 OK

Server: nginx/1.15.8

Date: Fri, 09 Apr 2021 13:34:03 GMT

Content-Type: text/html

Content-Length: 98

> blank space

<html>

<head>

<title> ToyHack Me </title>

</head>

<body>

welcome to ToyHack Me . com

</body>

</html>

To break down each line of response:

link 1:- HTTP 1.1 is the version of HTTP protocol the server is using and then followed by HTTP status code in the case "200 OK" which tells request has completed successfully.

link 2:- This tells us the web server software and version number.

link 3:- The current date, time and timezone of web server

link 4:- The content type header tells client what sort of information is going to be sent, such as HTML.

link 5:- Content length tells client how long the response is, this way we confirm no data is missing.

link 6:- HTTP response contains a blank line to confirm end of HTTP response.

link 7-14:- The information that has been requested, in this instance of home page

Qwiz:-

Q1) What HTTP protocol is been used in above example?

HTTP/1.1

Q2) What response header tells the browser how much data to download?

/ super :

A) content - length